

Maitreyi Swaroop — Curriculum Vitae

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EDUCATION

Carnegie Mellon University, Pittsburgh, USA

August 2024 – Present

Ph.D. Candidate, Machine Learning — Supervisor: Prof. Bryan Wilder

Department of Machine Learning, School of Computer Science

Indian Institute of Technology, Kharagpur, India

July 2019 – May 2024

BSc. (Hons) + MSc. in Mathematics & Computing

Department Rank: 2/63

Micro Specialization in Artificial Intelligence and Applications

Overall GPA: 9.45/10.00

RESEARCH EXPERIENCE & PUBLICATIONS

Distributionally Robust Feature Selection, *Maitreyi Swaroop, Tamar Krishnamurthi, Bryan Wilder.*

- **Paper accepted at NeurIPS 2025 Conference, Main Track (Poster)** [Code]
- Novel method for variable subset selection in a group-DRO setting. Project funded by the National Institute of Mental Health (NIMH).

Research Internship — PI: Amit Sharma, Vineeth Balasubramanian

Microsoft Research, Bangalore

June 2025 – August 2025

- Developed *monitor-guided reasoning*: a model-agnostic framework that improves language model performance through integration of a verification monitor into the generation loop.

Research Internship — PI: Prof. Dhanya Sridhar

Mila - Quebec AI Institute, Montreal, Canada

May 2023 – May 2024

- **Paper: Learning Macro Variables with Auto-encoders**, *Maitreyi Swaroop, Eric Elmoznino, Dhanya Sridhar.* Accepted at *NeurIPS 2023 Workshop on Causal Representation Learning*. [Code]
- Proposed **DeepCFL**: a self-supervised method that learns macro variables and their relations, extending the desiderata of Causal Feature Learning.

Masters Thesis — PI: Prof. Partha Pratim Chakrabarti

Indian Institute of Technology, Kharagpur, India

May 2023 – May 2024

- Developed hybrid methods combining approximation algorithms, heuristics, and RL approaches for routing problems, including a novel divide-and-conquer approach using RL models with Lin Kernighan k -opt solver.
- Awarded *Best Project Work* for my thesis.

Research Internship — PI: Dr. Manuel Gomez Rodriguez

Max Planck Institute for Software Systems, Kaiserslautern, Germany

May 2022 – July 2022

- Formulated a threshold test incorporating intersectionality of traits to identify biased decision-makers using Dirichlet-Process Mixture Models (DPMM) and MCMC inference methods.

ADDITIONAL EXPERIENCE & PROJECTS

Google Research Week — Google Research, Bangalore, India

February 2024

IIT Delhi Theoretical Computer Science Winter School — Indian Institute of Technology Delhi December 2022

Data Science Textbook Collaboration

December 2020 – May 2021

Indian Institute of Science (IISc.), Bangalore (Remote)

- Collaborated with Prof. Ramesh Hariharan and Prof. Rajesh Sundaresan on developing solutions for their Data Science textbook, focusing on astronomical calculations to determine Earth and Mars orbital parameters using Tycho Brahe's historical dataset—the same data employed by Kepler in formulating his laws of planetary motion.

AWARDS & HONORS

- **Best Project Work** — Awarded for the best Masters Thesis Project in the Department of Mathematics & Computing, 2024.
- **Indian Academy of Sciences (IASc) Summer Research Fellowship** — Awarded the IASc-INSANA-SASI Summer Research Fellowship in 2021 for a two-month research project at the Indian Institute of Science Education & Research (IISER), Kolkata. Could not participate due to the pandemic.